Cloud Sync APIs

Cloud Manager 3.8

Ben Cammett
October 30, 2020

This PDF was generated from https://docs.netapp.com/us-en/occm38/api_sync.html on May 31, 2021.
Always check docs.netapp.com for the latest.

Table of Contents

Cloud Sync APIs . . . 1
Getting started . . . 1
API reference . . . 2
Using list APIs . . . 2

Cloud Sync APIs

The Cloud Sync capabilities that are available through the web UI are also available through RESTful APIs.

Getting started

To get started with the Cloud Sync APIs, you need to obtain a user token and your Cloud Central account ID. You'll need to add the token and account ID to the Authorization header when making API calls.

Steps

1. Obtain a user token from NetApp Cloud Central.

```
POST https://netapp-cloud-account.auth0.com/oauth/token
Header: Content-Type: application/json
Body:
{
                "username": "<user_email>",
                "scope": "profile",
                "audience": "https://api.cloud.netapp.com",
                "client_id": "UaVhOIXMWQs5i1WdDxauXe5Mqkb34NJQ",
                "grant_type": "password",
                "password": "<user_password>"
}
```

2. Obtain your Cloud Central account ID.

```
GET https://cloudsync.netapp.com/api/accounts
Headers: Authorization: Bearer <user_token>
Content-Type: application/json
```

This API will return a response like the following:

```
[
  {
    "accountId": "account-JeL97Ry3",
    "name": "Test"
  }
]
```

3. Add the user token and account ID in the Authorization header of each API call.

Example

The following example shows an API call to create a data broker in Microsoft Azure. You would simply replace <user_token> and <accountId> with the token and ID that you obtained in the previous steps.

```
POST https://cloudsync.netapp.com/api/data-brokers
Headers: Authorization: Bearer <user_token>
Content-Type: application/json
x-account-id: <accountId>
Body: { "name": "databroker1", "type": "AZURE" }
```

What should I do when the token expires?

The user token from NetApp Cloud Central has an expiration date. To refresh the token, you need to call the API from step 1 again.

The API response includes an "expires_in" field that states when the token expires.

API reference

Documentation for each Cloud Sync API is available from NetApp Cloud Central.

Using list APIs

List APIs are asynchronous APIs, so the result does not return immediately (for example: `GET /data-brokers/{id}/list-nfs-export-folders` and `GET /data-brokers/{id}/list-s3-buckets`). The only response from the server is HTTP status 202. To get the actual result, you must use the `GET /messages/client` API.

Steps

1. Call the list API that you want to use.
2. Use the `GET /messages/client` API to view the result of the operation.
3. Use the same API by appending it with the ID that you just received: `GET http://cloudsync.netapp.com/api/messages/client?last=<id_from_step_2>`

   Note that the ID changes each time that you call the `GET /messages/client` API.

Example

When you call the `list-s3-buckets` API, a result is not immediately returned:

```
GET http://cloudsync.netapp.com/api/data-brokers/<data-broker-id>/list-s3-buckets
Headers: Authorization: Bearer <user_token>
Content-Type: application/json
x-account-id: <accountId>
```

The result is HTTP status code 202, which means the message was accepted, but was not processed yet.

To get the result of the operation, you need to use the following API:

```
GET http://cloudsync.netapp.com/api/messages/client
Headers: Authorization: Bearer <user_token>
Content-Type: application/json
x-account-id: <accountId>
```

The result is an array with one object that includes an ID field. The ID field represents the last message that the server sent. For example:

```
[
    {
        "header": {
            "requestId": "init",
            "clientId": "init",
            "agentId": "init"
        },
        "payload": {
            "init": {}
        },
        "id": "5801"
    }
]
```

You would now make the following API call using the ID that you just received:

```
GET http://cloudsync.netapp.com/api/messages/client?last=<id_from_step_2>
Headers: Authorization: Bearer <user_token>
Content-Type: application/json
x-account-id: <accountId>
```

The result is an array of messages. Inside each message is a payload object, which consists of the name of the operation (as key) and its result (as value). For example:

```
[
    {
        "payload": {
            "list-s3-buckets": [
                {
                    "tags": [
                        {
                            "Value": "100$",
                            "Key": "price"
                        }
                    ],
                    "region": {
                        "displayName": "US West (Oregon)",
                        "name": "us-west-2"
                    },
                    "name": "small"
                }
            ]
        },
        "header": {
            "requestId": "f687ac55-2f0c-40e3-9fa6-57fb8c4094a3",
            "clientId": "5beb032f548e6e35f4ed1ba9",
            "agentId": "5bed61f4489fb04e34a9aac6"
        },
        "id": "5802"
    }
]
```

Copyright Information

Copyright © 2021 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.